

Self-Concordant Perturbations for Linear Bandits

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Linear Bandits

Given an horizon n , an action set $K \subset \mathbb{R}^d$, and an unknown sequence of loss vectors $(y_t)_{t=1}^n$, sequentially choose an action $a_t \in K$ then observe punctual loss $\langle y_t, a_t \rangle$.

We define the *regret*:

$$R_n := \mathbb{E} \left[\sum_{t=1}^n \langle y_t, a_t \rangle \right] - \inf_{u \in K} \sum_{t=1}^n \langle y_t, u \rangle$$

FTRL & FTPL

FTRL: $a_t = \arg \min_{a \in K} \{ \sum_{s < t} \langle a, y_s \rangle + \psi(a) \}$

FTPL: $a_t = \arg \min_{a \in K} \{ \sum_{s < t} \langle a, y_s \rangle + \langle a, \xi_t \rangle \}$ with $\xi_t \sim D$

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- ▶ When $(y_s)_{s < t}$ are known, both part of the GBPA framework [Abe+14]

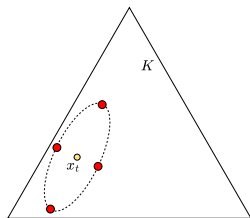
$$a_t = \nabla \Phi \left(- \sum_{s < t} y_s \right)$$

- ▶ Issue: we do not know the y_s . Instead, use estimates $\hat{y}_s$ of y_s using only $\langle y_s, a_s \rangle$. Need *randomization* when taking our action.

Exploration

- For FTRL-based alg., need for *additional exploration mechanism*

SCRiBLE [Abe+08]: x_t given by FTRL,
then sample $a_t \sim \mathcal{U}(\text{poles of } W(x_t))$

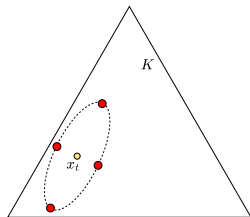


- Uses self-concordant barriers
- Worst-case regret in $O(d^{3/2}\sqrt{n \ln n})$

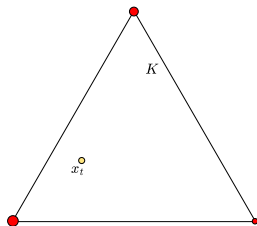
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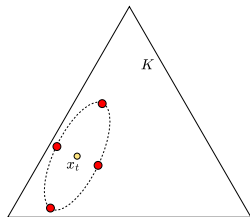


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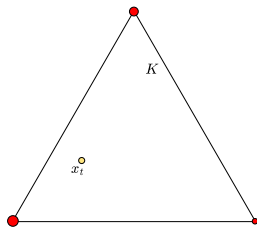
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Question: Can we get optimal regret using self-concordance and no additional exploration mechanism using FTPL ?

SC-FTPL

A distribution $\mathcal{D}$ is a ϑ -self-concordant perturbation for K if

$$\nabla \mathcal{R}^*(\theta) = \mathbb{E}_{\xi \sim \mathcal{D}}[\nabla \phi_K(\theta + \xi)], \quad \forall \theta \in \mathbb{R}^d$$

with $\mathcal{R}$ a ϑ -self-conc. barrier and ϕ_K the support function of K .

SC-FTPL: Given a ϑ -self-concordant perturbation $\mathcal{D}$, for all t ,

- ▶ Sample $\xi_t \sim \mathcal{D}$
- ▶ Play $a_t = \arg \min_{a \in K} \langle a, \hat{Y}_{t-1} - \frac{1}{\eta} \xi_t \rangle$
- ▶ Estimate $\hat{y}_t = \langle y_t, a_t \rangle Q_t^{-1} a_t$, where $Q_t = \mathbb{E}_{t-1}[a_t a_t^\top]$

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Lemma (Th. 5): If $2\eta \|\hat{y}_t\|_t \leq 1$ a.s, then

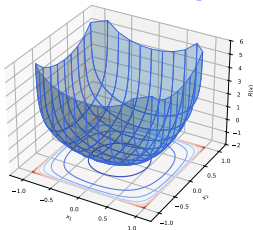
$$R_n \leq \frac{\vartheta \ln n}{\eta} + \eta \sum_{t=1}^n \mathbb{E} \|\hat{y}_t\|_t^2 + 2$$

- ▶ For SCRiBLLe $\mathbb{E} \|\hat{y}_t\|_t^2 \leq d^2$, yields $R_n = O(d \sqrt{\vartheta n \ln n})$

Hypercube

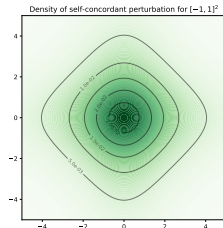
Entropic barrier

$\mathcal{R}^* : \theta \mapsto \ln \int_K e^{\langle \theta, x \rangle} dx$
 d -self-concordant on $[-1, 1]^d$



Replicating d -self-concordant
perturbation with density

$$f(x) = \prod_{i=1}^d \left(\frac{1}{2x_i^2} - \frac{1}{2 \sinh(x_i)^2} \right)$$

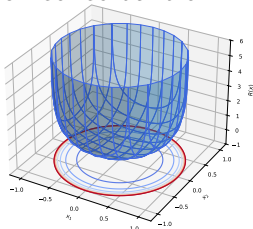


- Bounds on the estimator $\|\hat{y}_t\|_t^2 \leq 3d$ and $\mathbb{E}\|\hat{y}_t\|_t^2 \leq \frac{1}{3}d$
- Improvement over SCRiBLE by a $3d$ factor
- **Optimal** regret rate

$$R_n \leq \frac{2}{\sqrt{3}} d \sqrt{n \ln n} + 2$$

Euclidean Ball

Log-barrier $\mathcal{R} : x \mapsto -\ln(1 - \|x\|^2)$
1-self-concordant on $\mathbb{B}^d$



Replicating 1-self-concordant
perturbation given by

$$\xi = \mathbf{N}/ZU$$

with $\mathbf{N} \sim \mathcal{N}(0, I_d)$, $Z \sim \chi_{d+1}$, and
 $U \sim \mathcal{U}([0, 1])$

- ▶ Bounds $\|\hat{y}_t\|_t^2 \leq d^2 \eta \|\hat{Y}_{t-1}\| + 4d^2$ and $\mathbb{E}\|\hat{y}_t\|_t^2 \leq d^2$
- ▶ No improvement over SCRiBL
- ▶ **Suboptimal** regret rate

$$R_n \leq 2d\sqrt{n \ln n} + 2 + O\left(\frac{\ln^3 n}{d}\right)$$

[Bub+12] achieved $O(\sqrt{dn \ln n})$ regret on the ℓ_2 ball

Open Questions

- ▶ Do self-concordant perturbations exist for every convex body $K \subset \mathbb{R}^d$? What regret rate would we obtain ?
- ▶ In particular, can we replicate the entropic barrier [Bub+15]: $\mathcal{R}^* : \theta \mapsto \ln \int_K e^{\langle \theta, x \rangle} dx$, which is a universal d -self-conc. barrier

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paper & slides



Thank you for your attention